

OREBRO, Sweden

WMO: 024320

Lat: 59.229N

Lon: 15.046E

Elev: 53

StdP: 100.69

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-14.0	-19.9	0.6	-17.4	-16.2	0.9	-12.8	12.2	6.3	11.0	5.2	1.6	210	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	27.4	18.2	25.7	17.5	23.8	16.7	19.7	24.9	18.6	23.4	17.6	21.9	3.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.0	21.5	16.9	12.1	20.6	15.9	11.3	19.7	56.3	25.0	53.0	23.3	49.8	21.9	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.6	7.6	DB	-20.5	29.7	4.2	2.1	-23.6	31.2	-26.0	32.4	-28.4	33.6	-31.5	35.1	(4)
(5)			WB	-20.8	21.3	4.1	1.5	-23.8	22.4	-26.2	23.2	-28.5	24.1	-31.5	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-2.3	-1.9	1.0	6.1	11.0	14.9	17.5	16.2	12.1	6.7	2.5	-0.7	(6)	
	DBStd	8.01	5.45	5.04	4.34	3.51	3.60	2.79	2.70	2.61	3.04	3.57	4.04	5.53	(7)	
	HDD10.0	1833	382	333	279	124	29	1	0	0	16	111	225	333	(8)	
	HDD18.3	4183	640	566	537	366	227	108	50	76	188	359	474	591	(9)	
	CDD10.0	730	0	0	0	8	62	148	231	193	78	10	1	0	(10)	
	CDD18.3	40	0	0	0	0	1	6	23	9	1	0	0	0	(11)	
	CDH23.3	473	0	0	0	1	29	87	248	106	2	0	0	0	(12)	
	CDH26.7	80	0	0	0	0	2	12	53	13	0	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.9	3.9	3.8	3.7	3.7	3.4	3.1	3.1	3.4	3.5	3.7	3.9	(14)	
(15) Precipitation	PrecAvg	690	48	42	32	39	49	67	88	75	66	66	63	56	(15)	
	PrecMax	921	90	83	55	111	97	137	206	177	162	160	138	113	(16)	
	PrecMin	560	11	14	7	7	13	9	31	14	14	18	22	12	(17)	
	PrecStd	81	22	19	12	25	24	28	42	41	39	32	32	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.2	14.5	21.0	26.4	28.4	30.2	28.1	23.3	16.2	12.2	10.2	(19)	
		MCWB	6.8	6.3	8.8	11.8	16.1	17.7	19.9	18.2	16.7	13.5	10.8	9.1	(20)	
	2%	DB	6.5	6.8	11.1	18.1	23.7	25.9	28.1	26.1	21.1	14.6	10.9	8.5	(21)	
		MCWB	5.4	5.2	6.7	10.9	14.6	16.9	18.8	18.0	15.6	12.5	9.5	7.5	(22)	
	5%	DB	5.2	5.5	8.8	15.4	21.2	23.7	26.2	24.2	19.2	13.4	9.8	7.2	(23)	
		MCWB	4.2	4.2	5.5	9.3	13.8	15.8	18.2	17.4	14.7	11.6	8.6	6.2	(24)	
	10%	DB	4.1	4.4	7.0	13.0	18.7	21.5	24.1	22.1	17.6	12.2	8.2	6.0	(25)	
		MCWB	3.2	3.2	4.5	7.7	12.3	14.7	17.4	16.4	14.1	10.6	7.3	5.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.8	9.1	13.5	17.8	19.2	21.6	20.7	17.7	14.4	11.2	9.4	(27)	
		MCDB	7.4	7.7	13.1	19.3	23.3	25.1	26.7	25.2	21.6	16.0	12.1	10.1	(28)	
	2%	WB	5.5	5.4	7.3	11.2	16.0	17.7	20.2	19.3	16.4	13.0	9.7	7.6	(29)	
		MCDB	6.2	6.3	10.2	16.5	21.0	23.5	25.8	23.5	19.9	14.2	10.5	8.2	(30)	
	5%	WB	4.3	4.4	5.9	9.9	14.3	16.6	19.2	18.2	15.4	12.0	8.5	6.3	(31)	
		MCDB	5.0	5.3	8.2	14.5	19.6	22.0	24.5	21.8	18.3	13.2	9.4	7.1	(32)	
	10%	WB	3.2	3.2	4.7	8.5	13.0	15.5	18.2	17.3	14.4	10.9	7.4	5.0	(33)	
		MCDB	3.9	4.1	6.5	12.1	18.1	20.2	22.9	20.9	17.0	12.1	8.1	5.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	7.9	10.1	10.8	10.4	10.2	10.3	9.4	7.0	4.9	5.2	(35)	
		MCDBR	4.9	5.4	10.3	14.7	15.5	14.7	14.2	13.7	11.9	7.9	6.2	5.0	(36)	
	5% WB	MCWBR	4.8	4.5	7.0	8.4	7.8	7.1	6.5	6.9	7.3	5.9	5.6	4.9	(37)	
		MCWBR	4.8	5.0	9.4	12.8	12.9	12.6	12.4	11.5	10.6	6.9	5.7	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.275	0.315	0.337	0.375	0.371	0.361	0.384	0.375	0.344	0.333	0.301	0.259	(40)	
		taud	2.219	2.293	2.353	2.315	2.395	2.444	2.399	2.419	2.495	2.429	2.284	2.150	(41)	
	Ebn at Noon		532	698	800	826	859	874	841	820	793	676	495	406	(42)	
		Edh at Noon	44	69	87	107	106	103	105	96	76	60	42	32	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.32	0.92	2.46	3.90	4.99	5.66	5.11	4.03	2.69	1.21	0.40	0.21	(44)	
	RadStd		0.05	0.17	0.29	0.45	0.47	0.35	0.59	0.44	0.35	0.21	0.09	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.66	N/A	N/A	N/A	N/A	N/A	N/A	-237	N/A	N/A
(47) Regional (58 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-135	-162	N/A	N/A

Nomenclature: See separate page